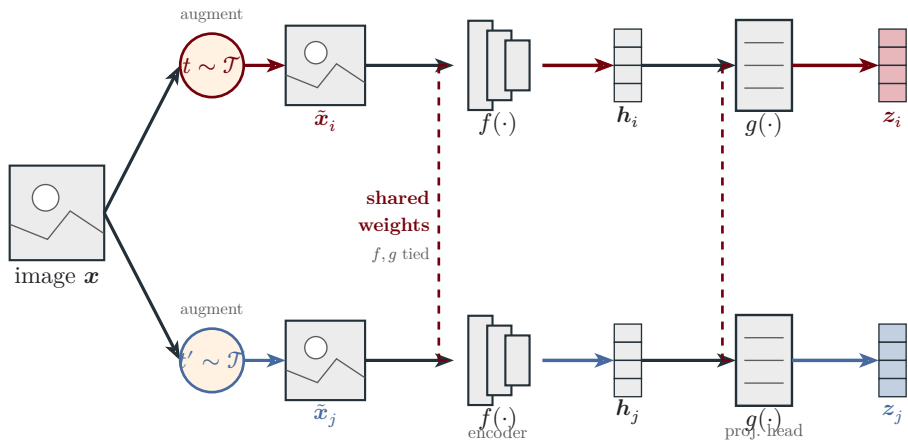


SimCLR — contrastive pretraining

two augmented views · shared encoder + projection head · NT-Xent in-batch contrast



NT-Xent loss

in-batch similarity $S_{ab} = z_a^\top z_b / (\|z_a\| \|z_b\|)$

candidate b

	$\tilde{x}_1^i$	$\tilde{x}_1^j$	$\tilde{x}_2^i$	$\tilde{x}_2^j$
anchor a $\tilde{x}_1^i$	—	+	—	—
$\tilde{x}_1^j$	+	—	—	—
$\tilde{x}_2^i$	—	—	—	+
$\tilde{x}_2^j$	—	—	+	—

■ positive same image
■ negative in-batch
□ self (masked)

NT-Xent (per positive pair i, j)

$$\mathcal{L}_{i,j} = -\log \frac{\exp(S_{ij}/\tau)}{\sum_{k \neq i} \exp(S_{ik}/\tau)}$$

numerator = positive · denominator sums the $2N - 1$ in-batch candidates temperature τ